

The Cryptocurrency “Pump-and-Dump”: Social Media and Legal and Ethical Ambiguity

Brady D. Lund

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Abstract

This chapter describes the dynamics of a crypto-trading “pump-and-dump” group – groups organized on Reddit and Discord channels that use social media to spread positive misinformation about a cryptocurrency in order to temporarily inflate its value and collect huge profits. It discusses the nature of cryptocurrency marketplaces, social networking related to crypto, and the pump-and-dump phenomenon, its social and economic impacts, and ethical concerns. Following the rise in the value of Bitcoin and the WallStreetBets/GameStop saga in December 2020 and January 2021, these pump-and-dump groups used the frenzy surrounding “get rich quick” investing to generate inordinate profits off of these ambitious individuals’ losses. Rallying around a shared philosophy and profit motive, these groups utilized social media disinformation campaigns to fool new crypto investors in squandering their funds, often while failing to acknowledge the ethical conundrum of stealing from the poor and ambitious. This study is the first to present a narrative of the cryptocurrency pump-and-dump schemes in the wake of the events of 2020/2021 that saw the frequency of these schemes escalate.

In January 2021, the “GameStop saga” became major news, while at the same time Bitcoin reached peak levels. It led to a peak frenzy of inexperienced investors who sought to “get rich quick,” and there were plenty of sharks wait to prey on the fresh herd. The GameStop trading bonanza was a once-in-a-lifetime event. Major hedge funds had bet heavy on the stock to deprecate in value. Some savvy investors took notice. When you short a stock, you borrow and sell a certain number of shares at a specific price, hoping then that the price of the stock drops such as that when it comes time to return the shares that you borrowed, you can purchase them at a lower price and pocket the difference as profit. When you short, though, you must always return what you borrow, regardless of where the price goes. So, if you short the entire quantity of GameStop shares in existence, and then a bunch of twenty-year olds with a Visa card buy up the shares in the meantime and inflate the price, the hedge funds are going to have to pay whatever inflated price those shares reach. Ultimately, the GameStop saga confirmed for many what they already thought: that game is rigged against them. The trading platforms barred the stock from being traded, causing the price to artificially drop and bailing out the hedge funds. However, these new investors had tasted the thrill of making serious profit for doing next-to-no work and, like true gambling addicts, many did not turn away completely from the game, but simply looked for looser slots.

Crypto tokens, or coins, are quantities of a digital assets (analogous to “shares” in a company), tracked on a ledger, that act as a form of completely-virtual currency (Narayanan et al., 2016). These tokens are created using blockchain technology: These “assets” are created through mining, a process whereby computer networks around the world compete to verify crypto transactions the fastest, consuming a lot of energy in the process (making crypto mining an unprofitable venture for many). Exchanges of crypto coins are tracked on an electronic ledger,

which is where the verification of transactions occurs. Records of crypto ownership are often stored in “wallets,” or a physical space (like a certain software) where the keys (code/password that identify ownership of an amount of crypto) are stored. However, many crypto traders today, particularly amateurs, simply allow the crypto marketplaces to store their currency for them (though this, of course, increases risks to privacy and security of the coin).

As the GameStop saga moved to a close, many of those who “got hooked” on the experience sought a new outlet to make large profits on short-term investments. Enter the world of crypto. Bitcoin had already nearly tripled in value in the month leading up to GameStop, and a robust market of thousands of cheaper crypto coins with seemingly boundless potential existed at the ends of everyone’s fingertips. Investing sites like Robinhood had made it possible to purchase crypto on their platforms (as easily as purchasing regular stock in company), but there are plenty of crypto-specific marketplaces in operation as well – with perhaps the most well-known being Coinbase (the eTrade of the cryptosphere). This marketplace is most people’s entry into the world of crypto investing, before the move on to other marketplaces with more diverse coin offerings (Coinbase is limited to only about 40 different coins). Traffic to these marketplaces grew, and simultaneously the Reddit groups for investing and crypto grew abuzz with activity. Opportunities seemed aplenty – and it was the perfect situation for the pump-and-dump groups to thrive.

What is a Cryptocurrency?

Cryptocurrency is a digital currency that is created and maintained (generally) using decentralized blockchain technology (Lee, Guo, & Wang, 2018; Narayanan et al., 2016). Cryptocurrency, in itself, is not anything physical. It is just a ledger of transactions that are validated by “miners,” who (in Bitcoin’s “proof of work” set-up) race to solve cryptographic

puzzle, with the solver having the right to verify blockchain transactions and, in exchange for their efforts, receiving compensation in the form of that cryptocurrency. When one purchases a crypto token, they are really purchasing a stake on that blockchain – not too dissimilar from purchasing stock in a company. However, unlike with stock shares, which represent an ownership share in an actual company, ownership in cryptocurrency (in most cases) does not represent ownership over any tangible asset. Because the value of cryptocurrency is not tied to any physical asset and instead relies on a belief among investors in its value, the cryptocurrency market can be volatile and susceptible to manipulation.

Legal Concerns with Cryptocurrency

Cryptocurrency itself poses several legal and ethical risks, as noted in recent legal and scholarly literature. Presently, cryptocurrency lacks any government oversight or regulation (Nestertsova-Sobakor et al., 2019). That is part of the intrigue of cryptocurrency for many investors: they want a new economy that is not based on, or susceptible to, government control (Vishwakarma, Khan, & Jain, 2018). Of course, governments themselves want to maintain economic controls – for reasons that are both magnanimous (economic insecurity threatens the livelihoods of the public) and narcissistic (it is more difficult to maintain power without some economic control). Governments are just now exploring ways to regulate cryptocurrency and proposed solutions may have taxation implications for cryptocurrency investors and legal ramifications for the operators of cryptocurrency marketplaces (Prytula et al., 2021).

One significant challenge with the legal regulation of cryptocurrencies is that new coins are being created constantly. If a government attempts to regulate a coin, investors can just switch to another. For this reason it is much more practical to regulate the marketplace – Binance, Coinbase, Crypto.com – than any currency itself (Inozemtsev, 2020). Enforcement of

this type of regulation, however, is inconsistent among countries (Drozd, Lazur, & Serbin, 2017; Cvetkova, 2018). In the United States, for instance, the main Binance trading platform (Binance.com) is blocked due to the United States' government's accusation that Binance was improperly using assets and attempting to avoid regulation (Khatri, 2020). In some countries though, like Pakistan and Ukraine, the sale of cryptocurrency is illegal, which means that these marketplaces are also banned (Bachynskyy & Radeiko, 2019; Grabowski, 2019). These approaches have additional own loopholes. New marketplaces can spring up in place of shuttered ones and cryptocurrency can be stored locally in "wallets" in order to avoid seizure of assets.

Given the desires of nations, and the vast amount of capital invested in cryptocurrency (over one trillion U.S. dollars, or more than 1/100 of the global annual GDP), it is likely that legal challenges will continue to arise and evolve. Just in the year 2021, China has enacted new restrictions on cryptocurrency and cryptomining (the process of validating cryptocurrency transactions in exchange for compensation in that currency) and India has aimed to follow suit (Kumar, 2021). The implications of cryptocurrency for international trade has yet to be fully explored. Given the fact that cryptocurrency transactions are impossible to trace, the potential for criminal activity is significant. One example of how cryptocurrency is used to support criminal activities is discussed in this chapter: the cryptocurrency pump and dump.

What is a Pump-and-Dump?

A pump-and-dump is a scheme where a relatively small group of people come together with the goal of defrauding a much larger group of an investment by manipulating the value of a stock (Huang & Cheng, 2015). They do this in (essentially) three steps: first, they purchase a large number of shares in the target stock at the lowest possible price; next, they hype up the stock on social media or email (or, historically, through word-of-mouth or phone calls) through

spam messages with disinformation about the potential of the stock (e.g., “it is going to jump 1000% overnight!”); finally, as people flock to purchase the stock and the price rises, the scammers sell off all their shares for a tremendous profit. Meanwhile, the new investors are purchasing those shares at the inflated rate, assuming that the value will continue to grow, only to find that it quickly plummets and leaves them with significant losses.

Pump-and-dump schemes are generally considered to be unlawful price manipulation (i.e., fraud), but are not heavily policed – in part because they are hard to police (Tillman & Indergaard, 2005). These are generally white-collar crimes (which makes the current developments in the cryptosphere even more fascinating, given that it is being done mostly by blue-collar and no-collar workers). It is difficult to conclusively prove what is intentional and malicious manipulation unless there is some kind of “smoking gun.” Instead, the market rules for trading were designed so that these schemes would be more difficult to pull off (at least until sites like Robinhood – which allowed for quick purchases of small amounts of shares – came around).

What makes cryptocurrency a good target for these pump-and-dump schemes? It lacks regulation almost completely. It is decentralized. The rules that govern how the crypto can be used, and how much will be created, are dictated by the owners of the crypto, not dissimilar from a stock holding. Most importantly, the craze over cryptocurrency, sparked mostly by the media coverage of Bitcoin, led to thousands of different crypto coins being created, and the market for these coins was and is still very volatile (Lee et al., 2018; Liu & Tsyvinski, 2018). 50% shifts in price from one day to the next are not uncommon, so pump-and-dump schemes are not difficult to pull off, nor are they too blatantly obvious as to raise serious negative publicity.

This chapter is not the first publication to discuss the cryptocurrency pump-and-dumps. There have been several technical papers on the topics, such as those by Kamps and Kleinburg (2018),

Mirtaheri et al. (2019), and Xu and Livshits (2020). Corbet et al. (2020) discuss hacking of crypto markets among other crypto cybercrime concerns. Unlike those other articles, though, this article presents a first-hand narrative of my experiences as an integrated member of one of these

pump-and-dump groups. **Mechanisms of Action: Reddit, Discord, and Binance**

It is like gambling mixed with social networking with a touch of “sticking it to the suckers.” Within Reddit, there are massive ecosystems of individuals interested in investing, both in stocks and in cryptocurrencies, and subreddits for topics like “Cryptopumping.” The major forums on Reddit are kind of like the Facebook wall of the pump-and-dump schemes. This is where the pumps are advertised after they have already started. The major forces behind the pumps have already bought in and use Reddit posts as their marketing platform to promote their pumps. Reddit does a fair bit of policing of posts, and many groups self-police as well. The real action behind a pump-and-dump generally occurs at a deeper level, not on Reddit but on Discord, a private messaging service. Telegram is another popular service that is used in the same way (Nizzoli et al., 2020). Users on Reddit will invite others to join their Discord group, or sometimes open invites will be posted in a Reddit forum. The following narrative is based on the author’s own experience as a member of a cryptopumping group.

In the typical medium-sized Discord, there are chats for stocks and crypto pumps, as well as a few long term investment projects. The long-term projects are selected because the group believed in the project’s mission and saw long-term investment potential. The term “100X” will be used to denote an investment that was seen as having the potential to increase in value by 100 times over the course of the next few years (generally, 2025 was seen as the “finish line”). Short-

term pumps are seen as ways to quickly increase the value of a stock or crypto by 100-200%, then quickly take those profits and invest in another pump. Members of the Discord will seek out cryptocurrencies that had a low or undervalued “market cap” – i.e., the total value of all of that crypto token that is owned. There are thousands of different crypto tokens in circulation that can be purchased from various marketplaces, so plenty of options existed. If a crypto has a market cap of \$10 million or so, this is a good spot, because it meant that it has sufficient investment to show that investors had interest but is low enough that the price could easily be manipulated.

To begin a pump, a large segment of the Discord will agree to a coin. These members will then purchase large quantities of that coin at its lowest cost. Price trends for the coin could be viewed on Coin Market Cap, which is kind of like the Yahoo Finance of crypto. Binance is a marketplace that is commonly used to purchase crypto, because it is widely-known and available and, unlike some crypto markets, it allows for stop-limit trading, where crypto can be bought or sold at a specified price after the market reaches another specific price. For instance, one might set a stop-price order to purchase \$1000 in Stellar (a popular crypto) at the price of \$0.45 per token when the market price of Stellar reaches \$0.44 per token. One would do this if they anticipated (but were not entirely sure) the market quickly rising, paying a premium for the token to ensure that the buy is processed before the price rises above the \$0.45 level.

After a coin is purchased, members of the Discord will take to social media – particularly Reddit, but also Twitter and YouTube – to advertise a pump, saying things like, “invest in Stellar now. Price is going to the moon!” These pumps were fairly successful in early 2021, because there were many uninitiated users on the major Reddit forums like r/WallStreetBets and r/Cryptocurrency, due to the media frenzy and social media promotion around the GameStop stock saga and the Dogecoin cryptocurrency promoted by Elon Musk. People who were late to

those two events but were intrigued by the idea of “getting rich quick” off of investing, flooded to Reddit in this wake. They could be convinced that, if they set up a Binance account and invested in a particular coin, they could see their profits “go to the moon” with a 10, 20, 100X growth. In reality, the growth, even in the biggest pumps, never reached that high. The biggest jump for Bitcoin was 4X, Dogecoin had a 10X jump for a brief moment, most others never got higher than a 300% jump. That level of increase over the course of a couple of days is nothing to scoff at, but it was not going to turn someone who invested \$100 into a millionaire overnight, as they may have thought.

Of course, many of those who invest never will see any increase at all. They were the marks. The people making the 300% increase are those from the Discord group who have invested before the pump began. The first few members of the pump from outside the Discord will also see good returns (if they dumped at the right time) but most people jump on the pump right as the Discord members are starting the “dump.” The dump is where the people jumping on the pump after seeing that the coin had gained 3X value in a day started being sold the coin of the Discord members, who were now raking in their profits. The skyrocketing price reaches its peak and starts to fall. Some investors will say it was just a small correction in the price before it would continue shooting up “to the moon.” They will only see their investment continue to drop in value. Those who invested at the 3X price, expecting it to rise to a 30X price, are now left with a coin worth only 1/3 of what they paid for it. This is how the game works. Stealing from the suckers, turning a \$100 investment into a \$200 profit, and then turning around and doing the same thing again.

This is not to say that there is no risk for the pump-and-dumpers. It is indeed like gambling because the pump does not always work. The members are not always successful in

convincing the average joes surfing Reddit to invest in their coin – if a batch has gotten burned enough times, they will turn away, and leave the pumpers waiting for the next batch to crop up. In these cases, pumpers have to turn on each other. Those who pull out fast enough can recoup their investment. Those who do not will take a loss on the coin. But there is always a next coin coming down the pike where one can earn it all back. Part of the fun is not just bringing in a profit: it is the comradery of being a member of this group where you could pull one over people, like information age street hustlers. One can make friends with a bunch of anonymous figures on the Internet, work together to pull off a heist, and then laugh about how big their profits were for a “day’s work.”

The Ethics of the Crypto Pump-and-Dump

Though I am a crypto investor, I did not directly participate in a pump-and-dump scheme, though I was, for several months in early 2021, a member of several pump-and-dump Discord channels and I would frequently participate in discussions held on these boards. I even recommended a coin once for a pump-and-dump that the group selected. These groups were intriguing not just because of what they do but because of how they are structured: with named leadership that had little actual control in a democratic system of suggesting and voting, but nonetheless held considerable sway over the decisions of the other group members. Many of the members felt that they have a righteous justification for their actions as a member of the group. As one anonymous member of a Reddit group posted,

We have all been inoculated with the power to change our future for the better.

Something that has been inherently taken from us in one form or another over the past year. Yet, while we sit at home or at work, we watch our world burn, our jobs taken, our freedoms slashed. We hear our governments rationalize with us as to

why they need more money from us and decide which businesses need to close.

While behind a thinly veiled curtain, they pump cash to the wealthiest, greed ridden corporations, and financial institutions all over the world.

At some point, though, it becomes evident what these pump-and-dump schemes really were: theft in broad daylight – a pyramid scheme, a con, a hustle, a crime. And this theft is not perpetrated against some wealthy banker in all likelihood, it is perpetrated against other common people who are looking to make a profit on a promising crypto coin. This seems antithetical to what the WallStreetBets/GameStop traders claimed as their mission, that they were fighting back against the millionaire and billionaire hedge fund managers who were destroying companies and other investors with their own brand of market manipulation. That is not what is being done with the pump-and-dump schemes. It is all about the profit and, while the group members often justify their actions, like in the example above, by suggesting that it is necessary to fight against a financial system that is rigged against them, that is not actually what they were doing – they are not fighting against the financial system, they are stealing from one another.

One time, I posted a comment in one of the chats about the ethics of the pump-and-dump approach, “if we know the price is going to drop, then we know someone must be losing money. Can we actually feel okay about the fact that we are making money by tricking others and taking it from them?” These types of questions never really appeal to the group, though. The philosophy is reserved for a few members/leaders who will ramble about it from time-to-time in order to motivate people – not unlike a football coach yelling at their players – but the average pump-and-dump investor seemingly has a fairly simple investing-mind, one that is trained to seek out the next opportunity to produce a profit. Some might say, “I’m investing my last \$100,” not realizing – or perhaps not caring – that another person saying the same thing just lost their last

\$100 as a victim to the scheme. Others will argue, “crypto pump-and-dumps aren’t illegal,” completely avoiding the true point of the question.

Nonetheless, there is a fair number of people who agree with my stance. There are “more ethical” groups that emerged that are “pump-and-hold” rather than pump-and-dump. The pump-and-hold name is a bit of a misnomer. These groups aim to create a pump that will inflate the price of a coin, but then only sell off about 25% of their coin at the inflated price, hoping that the overall price of the coin will drop only a little and then stabilize at a new elevated level, at which point they can sell off the remainder of the coin without feeling that they were swindling people. For instance, one might buy Stellar at \$0.23, inflate the price to \$0.71, sell off 25% of their balance at this profit level, have the price drop to \$0.57 and then stabilize, at which point they can sell off a good chunk of their remaining bounty. These pump-and-hold schemes are perhaps not actually ethically superior to the pump-and-dump ones, but they do make the members of these groups feel better about their actions, which is all that is important, pragmatically, for these schemes to continue.

A lesser (perhaps) ethical concern is the use of the ill-gotten funds by some members of the group. Crypto is the currency of choice for many of the hidden services available on the “dark web.” Much like with crypto, there is positive potential with the dark web, certainly for privacy purposes, but the idea of members advertising their use of their profits on illegal services, on purchasing pornography, etc. on these forums with no rebuke seems problematic, if expected.

The Legality of Pump-and-Dump Schemes

The legality of pump-and-dump operations can vary from country to country (Siering, 2019). In most developed countries it is prohibited to participate in these schemes, though it is

difficult to police. Authorities would need to uncover the identities of members of pump-and-dump groups and then also supply evidence that they actually participated in the scheme. One of the most notable cases where participants in a pump-and-dump scheme were prosecuted was the Enron scandal at the start of the 21st century. In this case, major investors and executives in the company participated in inflating the stock price of Enron and then selling off their shares before the price cratered. At the end of the year 2000, the average price for a share of Enron stock was over \$80. By the Fall of 2001, the average price was under \$1. The fraud perpetrated to sustain this operation was discovered by authorities in late 2001 (Chambers, 2002). Because of the bureaucracy in place within the large organization, there was a substantial trail of evidence. This type of trail is less evident in the case of the cryptocurrency pump-and-dump, as the communication is more decentralized and the target (cryptocurrency instead of stocks) is far less regulated.

In the United States, there are several laws related to securities fraud that have been used to prosecute pump-and-dump schemes. The Securities Act of 1933 and the Securities Exchange Act of 1934 allow the Securities Exchange Commission to prosecute any attempts to defraud or manipulate market securities. “Securities” here tends to be the key word that cryptocurrency pump-and-dumpers use to justify the legalities of their schemes. A security is just a representation of an ownership/position, while cryptocurrency not representative of a physical ownership in anything. There have not really been any specific laws passed to regulate the cryptocurrency-type markets that trade a real commodity but do so at a rapid rate like a stock exchange.

According to Strickler (2021), the type of activity that occurred with the GameStop saga and other pump and dump schemes could qualify as price manipulation under the Securities

Exchange Act of 1934, where manipulation is defined as, “a series of transactions to induce the purchase or sale of a security by others” (para. 3). However, Strickler concludes that there is likely insufficient evidence to prosecute the GameStop investors under the current law, since the law was designed to protect individuals from institutions (not the other way around) and there is little evidence that the investors intended to deceive anyone (they did the pump but were very clear and open about it). The case is somewhat different for crypto pump and dumpers, since they do intend to deceive. However, the existing law is still too ambiguous to adequately address the prosecution of crypto pump and dumpers to where crackdowns are feasible. Nations that want to pursue enforcement must consider enacting specific legislation that addresses the unique asset that is cryptocurrency.

Conclusion

In many ways, the democratization of online investing has been a tremendous boon to the average person. A low-wage worker can download an app like Binance or Robinhood and invest just a few dollars in a high-risk/high-reward crypto token that may, within days, double or triple in value. However, this type of investment is not really investing at all. It is not much more than gambling – selecting a coin and hoping that its price will jump (Liu & Tsyvinski, 2018). In trying to gain an advantage in this system by jumping on a supposedly “hot” coin, many investors are losing non-trivial amounts of money as victims to a culture of pump-and-dump schemers.

While it would be easy to write-off the schemers as common con-artists, it is evident that beyond the scheming, participation in these groups is a point of social connection. There is a sense of freedom in the level of anonymity that can be attained on the social media sites like Reddit and Discord, and the idea of democratically participating in the system of debating,

selecting, and promoting a coin, while also pocketing a profit, is intriguing to many. They have a sense of morality, but it is a sense of morality that may not be too distinct from that of a street criminal, that it is okay to engage in these schemes because you have been economically repressed. That one can connect with others that feel the same way, and together earn a considerable profit, is liberating. After all, are they not simply doing the same types of things that Wall Street investors do every day, at much higher levels and to much greater harm? That was the whole idea behind the GameStop and WallStreetBets saga: major hedge funds had bet on and were working to accelerate the company (and its investors') demise, but the common man took notice and fought back, making the hedge funds pay dearly. However, the difference between the GameStop saga and the pump-and-dump crypto schemes is that the latter targets the average man, not the billion dollar hedge funds. It turns the romantic, robin hood-esque idea of taking from the rich and giving to the poor into a perversion where the poor are simply taking from the more-oblivious poor.

The truly fascinating aspect of this saga is that it is difficult to ever stop the cycle of pump-and-dumps from continuing. Participants utilize a number of fake accounts, messages among pump-and-dump groups occur on private channels, and the objects of the exchange were purposely designed to preserve anonymity as they are exchanged. As long as there are suckers who think they can "get rich quick" off of crypto, there will be groups that will take advantage of them. Nonetheless, the illegality of these schemes, and the severe penalties handed out for past pump-and-dump offenses, should serve as a warning to those who would consider participating in one. Awareness of these schemes and investment/financial literacy for new crypto investors, as well as appeals to the morals of these schemers, can make a quick, tangible impact. While cryptocurrency has exciting potential and investing in these coins in a marketplace can be a fun

and profitable activity, these bad actors threaten to cause irreparable harm to the legitimacy of the crypto ecosystem in the eyes of the public.

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